June 2026

GSA NEXT BPA — Support Area 5
Artificial Intelligence Strategy & Implementation

Technical Factor 3 — Oral Presentation

Capital Technology Group (CTG) + Data Surge · GSA Schedule CTA

Agenda

1	Team CTG	CORPORATE REP
2	FALCON AI/ML Lifecycle Walkthrough	AI/ML ENGINEER
3	Use-Case Suitability & Deployment Readiness	AI STRATEGIST
4	Model Governance & Risk Management	AI GOVERNANCE LEAD
5	Securing AI, Interoperability & ATO Integration	SOLUTIONS ARCHITECT
6	Staffing Model & Why CTG	PRODUCT MANAGER

~60 minutes · five presenters · time reserved for your questions and any scenario

FALCON AI/ML capability — full lifecycle walkthrough

Problem framing:
USCIS needed identity matching across heterogeneous Federal source systems where biographic data was inconsistent — generating manual reconciliation backlogs and verification delays.

Data preparation:
Versioned training/evaluation datasets in Amazon S3; profiled for completeness, freshness, schema drift, and label quality. Privacy enforced in code via Pydantic data models.

Model selection & validation:
XGBoost and ruleset ensemble, optimized for latency and interpretability. Validated against labeled golden set of hard-case examples. F-β improved from 0.43 to 0.92 vs. legacy service.

Deployment & monitoring:
Containerized via Docker on Amazon EKS, fronted by Apigee. Training on SageMaker with metric-gated promotion. Databricks + New Relic + Splunk for drift, calibration, and SLO tracking.

0.91 precision vs. 0.40 legacy

0.93 F1 score vs. 0.54 legacy

5.8M+ users protected

DEMO

Production AI running at USCIS today — not a proof of concept.

Model boundaries & human oversight — what FALCON does and doesn't do

Match Model

Routes identity matches across SAVE & E-Verify. Never auto-adjudicates — non-matched cases and those requiring scrutiny route to human review.

ARGOS

LLM-based E-Verify enrollment fraud detection. USCIS investigators review every flagged registration before any enforcement action.

AI Governance Board

Both models completed the Risk Posture Review and AI inventory required by the USCIS AI Governance Board, with formal sign-off.

Key principle

No model auto-adjudicates an individual. Human-in-the-loop is enforced in code and in governance — not a feature to be added later.

ARGOS and Match Model both carry formal AI Governance Board sign-off. Certs: CDMP, CISSP on the governance team.

Use-Case Suitability Assessment — when AI is (and isn't) the answer

Our Use-Case Suitability Assessment covers five gates before we recommend AI.

- Is the outcome measurable and defined?
- Is sufficient labeled data available?
- Does the decision tempo favor ML over rules?
- Does the benefit justify complexity and risk?
- Has governance formally approved the use case?

We have recommended against AI when rule-based logic was more accurate and less brittle — and can walk through specific examples. Traditional approaches win when data is sparse or decision logic is simple.

1. **Defined outcome**
2. **Data available**
3. **Tempo fits**
4. **Benefit justifies**
5. **Approved**

We open every engagement asking: should AI even be the answer?

Deployment readiness — our eight-point assessment

Data & Governance

GATE 1-2

Data availability, permissions, label quality, and PII handling confirmed. Approved access paths documented.

Tooling & Access

GATE 3-4

Approved compute, APIs, model training environments, and access paths in place.

Ownership & Support

GATE 5-6

Accountable owner named. Technical support, training plan, and incident runbook assigned.

Telemetry & Readiness

GATE 7-8

Monitoring, alerting thresholds, governance path, and operational readiness criteria documented before launch.

Eight gates before a use case goes to production.

Bounded pilots & experiments — how we define success, then decide

"The pipeline carries the security burden so the engineering team can focus on the mission."

1 Define success up front

Metric thresholds, workflow scope, evaluation criteria, and the evidence needed to refine/scale/stop — all documented before the pilot runs.

2 Run bounded and observable

Pilots run against a defined scope with full telemetry. No "we'll know it when we see it" — every outcome is measurable.

3 Evidence-based decision

Refine if the model underperforms within scope. Scale if it clears thresholds. Stop if the use case is not ready — document why.

We define success criteria before we run a pilot — not after.

Model governance framework — documentation, drift & audit trails

DOCUMENT

- Model cards: purpose, training data, evaluation metrics, known limitations.
- Decision log for each major model choice.
- AI-specific risks documented in SSP under NIST 800-53 rev 5.

MONITOR

Ongoing:

- Drift detection on precision, recall, F-β vs. golden set.
- Calibration shifts and SLO tracking in New Relic → Splunk.
- Automated alerts trigger retraining runbook.

RETRAIN & AUDIT

Quarterly:

- Quarterly retraining on Amazon SageMaker per written SOP.
- Metric-gated promotion — model cannot advance without clearing all thresholds.
- Every inference recorded to Aurora prediction log for audit and attribution.

Governance is continuous — not a one-time review before launch.

AI risk management — NIST AI RMF alignment

Bias	GOVERN · MAP	Golden-set evaluation includes hard-case examples as bias stand-in. Human-in-the-loop prevents automated adjudication on any individual.
Hallucination	MEASURE · MANAGE	LLM outputs (ARGOS) constrained to classification with confidence thresholds. Human review required before any enforcement action.
Data poisoning	MAP · MANAGE	Training data versioned and hash-pinned. Provenance tracked from source to artifact. SBOM attested at every build.
Adversarial input	MEASURE · MANAGE	Inference inputs filtered through Pydantic schemas — only required fields reach the model. Anomalous input patterns trigger alerting and IR-4/IR-5 incident response.

We map every AI risk to a NIST AI RMF function before deployment.

Securing AI systems — access, logging & incident response

Access Control

- RBAC scopes human access to training data, model artifacts, and inference APIs by role.
- ABAC gates workload-to-workload authorization via Pydantic-filtered inference inputs.
- Only non-PII fields reach the model; response objects never expose PII to callers.
- PIV/CAC enforced for high-impact workloads. HashiCorp Vault for short-lived dynamic credentials.

Logging, Monitoring & IR

- Every inference recorded to Aurora prediction log for audit, drift detection, and per-decision attribution.
- New Relic + Splunk: performance SLOs, drift alerts, and security event correlation.
- AI incidents follow IR-4/IR-5 — ISSO engaged on anomalous model behavior.
- ConMon extended to AI telemetry: performance fidelity, version provenance, and drift.

AI components are governed under the same ATO boundary as the rest of FALCON — not a shadow system. Every AI risk is documented in the SSP.

AI components inherit the same security posture as the rest of the platform.

Interoperability, portability & avoiding vendor lock-in

Open standards	REST APIs, open data formats (JSON, Parquet), containerized model serving — no proprietary runtimes required.
Exportable models	Model artifacts in standard formats, such as pickle binaries. Training code, evaluation scripts, and SOPs are Government-owned IP from day one.
Data sovereignty	All training data and prediction logs remain in Government-controlled infrastructure in common structures: Databricks and PostgreSQL tables, S3 objects. No vendor retains model training data.
Transition support	Documented runbooks, retraining SOPs, and Government staff trained to run retraining cycles independently before project closeout.

The Government retains full data and model ownership — by design.

Model lifecycle management — versions, regressions & bias

1

Version control

Model artifacts, configs, and training code versioned in Git alongside the application. Any prediction can be traced to the exact training run and data version that produced it.

2

Metric-gated promotion

A model version cannot advance to production without clearing all acceptance thresholds — precision, recall, F-β, and bias metrics reviewed at each gate.

3

Regression & bias detection

Each retraining run compares against the incumbent on the same golden set, including hard-case examples. A regression on bias metrics blocks promotion regardless of overall accuracy.

Model version, training data hash, evaluation metrics, and human approval are all recorded before any production promotion.

Model changes are managed like code changes — with gates and evidence.

Human oversight, traceability & auditable AI operation

1 Source attribution

Every inference traced to input data, model version, and training dataset.

2 Prediction log

Aurora log records each inference for audit, drift detection, and per-decision review.

3 Human review gate

Match Model and ARGOS route uncertain or high-impact decisions to human review before action.

4 Model card

Published purpose, limitations, evaluation metrics, and known failure modes for every model.

5 Governance Board

Formal AI Governance Board sign-off at Risk Posture Review before production deployment.

6 Continuous monitoring

Performance fidelity, version provenance, and drift tracked post-authorization under CA-7.

Every AI decision is traceable — source, model version, and human reviewer.

AI in the ATO process — security, PII & continuous monitoring

AI-specific SSP

AI risks documented in the System Security Plan with NIST 800-53 rev 5 controls. Model artifacts and parameter configs version-controlled and referenced in the SSP.

Data provenance & PII

PII kept out of inference paths via Pydantic schemas. Only non-PII fields logged. Tests run against synthesized or masked data.

ATO boundary

FALCON AI components operate under USCIS AO authorization on FedRAMP-authorized AWS. AI is not a shadow system — it inherits the same authorization boundary.

Continuous monitoring

CA-7 continuous monitoring extended to AI telemetry: performance fidelity, version provenance, drift, and SLO thresholds reported to the AO.

AI model monitoring is integrated into the same ConMon program as the rest of the platform. Certs: CDMP, CISSP, CISM on the AI governance team.

AI components live inside the ATO boundary — not adjacent to it.

Staffing model & knowledge transfer — AI engagements

PROGRAM MANAGER

Core AI team:
Data Scientist, ML Engineer, MLOps Engineer, Domain SME, AI Governance Lead, Solutions Architect — six roles embedded per engagement.

Domain SME balance:
Domain SMEs embedded from day one, not added at demo time. Business logic cannot be learned from data alone.

Knowledge transfer:
Government staff trained on retraining SOPs, monitoring dashboards, and model card maintenance throughout the engagement — not in a final handoff sprint.

Dependency reduction:
Exit criterion: Government staff successfully runs one retraining cycle end-to-end before project close-out.

6	core AI roles per engagement
Day 1	domain SME integration (not end-of-sprint)
100%	knowledge transfer target before closeout

TOPIC 14

The Government can run model retraining independently by project close-out.

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Why Team CTG

AI in production, not demos

FALCON Match Model and ARGOS run at USCIS today — 5.8M+ E-Verify users, formal AI Governance Board sign-off, live ATO.

Governance built in

NIST AI RMF, model cards, metric-gated promotion, and human-in-the-loop are structural — not checkboxes added at launch.

Government retains control

Open formats, Government-owned IP, and knowledge transfer that ends dependency — by design.

Ready to walk through the live system. We welcome your questions and any scenario.

Appendix

Backup material for Q&A and scenario-based assessment

Scenario — model performance degrades post-deployment

1	**Alert triggered**	Drift alert fires in New Relic when precision drops below threshold on the golden-set evaluation.
2	**Investigate**	ML Engineer pulls prediction log from Aurora; Data Scientist reviews calibration shifts and hard-case failures.
3	**Assess impact**	AI Governance Lead documents residual risk; ISSO advises AO on operational impact and authorization status.
4	**Retrain or roll back**	Retrain per SOP on updated data — metric-gated promotion required before return to production. If urgent: roll back to prior version.
5	**Document & close**	Model card updated; retraining decision and evidence logged to audit trail; AO notified; ConMon record updated.

Governance holds — even under pressure.